Siqi Chen

Beijing, China chensq23@mails.tsinghua.edu.cn +86 18811671195 jamescsq47

Education

Tsinghua University, undergraduate *Aug 2023 - Present*
BS in Electronic Engineering

- **GPA: 3.9/4.0** (Rank: 10/261)

Massachusetts Institute of Technology, intern *July 2026 - Sep 2026*

- Research assistance at HANLab.

Tsinghua University High School, graduate *Aug 2017 - Jun 2023*
High School certificate

- College entrance exam: 700/750

Research Experience

My research interests focus on algorithm-system co-design, machine learning systems, and efficient deep learning algorithms. My previous research focused on accelerating inference for different models from both algorithmic and systemic perspectives.

Visual Generation Model *Feb 2025 - Dec 2025*

- **PAROAttention(NeurIPS 2025)** designs a novel Pattern-Aware token ReOrdering (PARO) technique, which unifies the diverse attention patterns into a hardware-friendly block-wise pattern. This unification substantially simplifies and enhances both sparsification and quantization
- **db-SP(MLSys 2026, first author)** systematically defines the dual-level (head-level and block-level) workload imbalance that arises from applying block-wise sparse attention with sequence parallelism methods. It proposes a dual-balanced workload partitioning approach and a sparsity-aware strategy selection mechanism to balance the workload at a negligible overhead.

Large Language Model *Jan 2026 - Present*

- **QLPO(COLM 2026)** proposes QLPO, a GRPO variant that favors short correct and long incorrect responses while preserving the correct/incorrect ratio during the rollout procedure of reinforcement learning to control the response length implicitly.
- **OARS(Under review at ICLR 2027)** uses a frozen large-model orchestrator to delegate subtasks to small-model workers that return concise reports. Workers are trained with contribution-aware rewards and hierarchical credit normalization to improve collaboration and reduce inference costs.
- **DualPD(Under review at MLSys 2027, first author)** improves PD-disaggregated agentic LLM serving through bidirectional transfers between prefill and decode workers, using direct GPU transfers or CPU staging according to tool readiness and resource availability. It also defers decode-side memory allocation until prefill completes.

Publications

BrainOmni: A Brain Foundation Model for Unified EEG and MEG Signals May 2025

Qinfan Xiao, Ziyun Cui, Chi Zhang, **Siqi Chen**, Wen Wu, Andrew Thwaites, Alexandra Woolgar, Bowen Zhou, Chao Zhang
neurips.cc/virtual/2025/poster/117066

PAROAttention: Pattern-Aware ReOrdering for Efficient Sparse and Quantized Attention in Visual Generation Models May 2025

Tianchen Zhao, Ke Hong, Xinhao Yang, Xuefeng Xiao, Huixia Li, Feng Ling, Ruiqi Xie, **Siqi Chen**, Hongyu Zhu, Yichong Zhang, Yu Wang
neurips.cc/virtual/2025/poster/117769

db-SP: Accelerating Sparse Attention for Visual Generative Models with Dual-Balanced Sequence Parallelism Oct 2025

Siqi Chen, Ke Hong, Tianchen Zhao, Ruiqi Xie, Zhenhua Zhu, Xudong Zhang, Yu Wang
https://mlsys.org/virtual/2026/oral/3798

QLPO: Quadrant-weighted Sampling for Length-aware Policy Optimization May 2026

Siwei Chen, **Siqi Chen**, Xupeng Miao, Bin Cui
https://openreview.net/forum?id=WHc6ywhpSZ

OARS: On Post-Training for Cost-Efficient Heterogeneous Multi-Agent Systems Oct 2026

Shang Yang, Zhichen Zeng, Shi Dong, Yichuan Wang, **Siqi Chen**, Junxian Guo, Joseph E. Gonzalez, Banghua Zhu, Song Han
Under review

DualPD: Bidirectional Dual-Path KV Transfer for Efficient Prefill–Decode Disaggregated Agentic LLM Serving Oct 2026

Siqi Chen, Shang Yang, Qinghao Hu, Junxian Guo, Song Han
Under review

Projects

An Intelligent Voice Assistant *Jul 2024 - Sep 2024*

- Complete an intelligent voice assistant which can remind users to check emails , finish regular tasks and focus on study
- Tools Used: Arduino, PCB, Python
- **Second Award of Hardware Design Competition**

Selected Honors

- Sensetime scholarship (2026, **Top 30 undergraduate AI researcher in China**)
- National Scholarship for Undergraduate Students (2024, 2025, **Highest honor for Chinese undergraduate**)
- Tsinghua Alumni - Dai Yusen Science and Technology Innovation Scholarship (2025, 2026)
- Tsinghua Alumni - Zhihua Integrated Circuit Scholarship (2024)
- First Prize, The 40th National College Student Physics Competition (2024)
- Comprehensive Scholarship of Tsinghua University (2024, 2025, 2026)
- Tsinghua University Five Star Volunteer (Highest Honor)
- Outstanding Student of Beijing (2020, 2023)

Technologies

Programming Languages: C++, C, Python, CUDA, LATEX

Software/Tools: MATLAB, Typst

Language: TOEFL:106 (30 29 22 25)